

EduVision_DV — Final Documentation

Higher Education Performance Dashboard

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1. Project Statement

EduVision_DV is a higher education analytics dashboard suite built to analyze university rankings, research performance, student diversity, and global education trends. Using publicly available datasets such as QS World University Rankings and Times Higher Education World University Rankings, raw educational data is transformed into actionable insights through an interactive Power BI workbook.

The dashboard suite enables students, academic researchers, university administrators, policymakers, and education consultants to evaluate institutional performance, compare universities globally, analyze research output, and identify trends in higher education.

2. Dataset Sources

Property	Value
File	university_cleaned.csv
Rows	22,125
Columns	38
Years covered	2016 – 2026
Sources	QS World University Rankings, Times Higher Education World University Rankings

Key data fields include university name, world rank, overall score, country, region, subject field, publications count, citations count, H-Index, citations per faculty, academic reputation score, research productivity index, total students, international students count, faculty count, gender split, undergraduate/postgraduate counts, and country-level average metrics.

3. KPI Definitions

KPI	Description
Top Global Ranking	Highest-ranked university under the selected filters.
Total Universities	Total number of universities represented in the dataset.
Average Overall Score	Average performance score across all universities.
International Students %	Percentage of students who are international.
Faculty-to-Student Ratio	Number of students per faculty member, shown as 1 : X.
Total Publications	Total research publications across universities.
Total Citations	Total citations received across universities.
Average H-Index	Average H-Index across universities.
Average Citations per Faculty	Average citation impact per faculty member.
Total Students	Total student population represented in the dataset.

KPI	Description
Gender Split (Female % / Male %)	Average gender distribution across universities.
Total Countries	Number of countries represented in the dataset.

4. Dashboard Guide

University Overview

- Top 10 university rankings
- Global university distribution by region
- Academic reputation analysis
- University performance trends over time
- Institutional comparison

Research Analytics

- Publications analysis
- Citation performance
- Research productivity trends
- Top research institutions
- Research impact comparison

Student Analytics

- International student analysis
- Faculty-to-student ratio analysis
- Student diversity trends
- Enrollment comparisons
- Student distribution analysis

Country Comparison

- Country ranking comparison
- Education performance benchmarking
- Regional education trends
- Top performing countries

5. Education Analytics Methodology

- University names and country names were standardized during data cleaning to ensure consistent grouping and comparison.
- KPIs were engineered using DAX in Power BI, built directly on the cleaned dataset.
- Global filters (Year, Region, Country, Subject Area) are synced across all four dashboards, allowing consistent, interactive exploration.
- Ranking and performance metrics were validated against the source dataset to ensure accuracy.

6. Tech Stack

Area	Tools
Data Collection	Python, QS Rankings, Times Higher Education Rankings
Data Processing	Pandas, NumPy
Data Cleaning	Python
Visualization	Power BI Desktop
Dashboard Integration	Power BI Filters, Slicers, Navigation
Documentation	Markdown, Word, GitHub

7. Conclusion

The EduVision_DV project successfully delivers a complete, interactive higher education analytics dashboard suite. Global university ranking and performance data was collected, cleaned, and transformed into a unified Power BI workbook consisting of four interconnected dashboards: University Overview, Research Analytics, Student Analytics, and Country Comparison.

All planned KPIs were engineered and validated, dashboard interactions and filters function as intended, and the workbook provides a clear, interactive way to explore university rankings, research performance, student demographics, and country-level education trends. The project meets its evaluation criteria — data completeness above 95%, KPI accuracy above 95%, all four dashboards fully integrated, and complete supporting documentation — making EduVision_DV a portfolio-ready, fully documented data visualization project.